MPG No. 21.3.78 6′ [Type III, hexagonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss).

$$\begin{bmatrix} M_{xxx} & M_{xxy} & 0 \\ -M_{xxx} & -M_{xxy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxy} & -M_{xxx} & 0 \end{bmatrix}$$

$M_{xxx} = \mathbb{M}_{3c}^{(1)}$
$M_{xxy} = \mathbb{M}_{3s}^{(1)}$

* Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xxyz} & -M_{xxyz} & 0 & 0 & 0 & -M_{xxzx} \\ M_{xxzx} & -M_{xxzx} & 0 & 0 & 0 & M_{xxyz} \\ 0 & 0 & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$M_{xxyz} = \mathbb{M}_{4s}^{(1)}$
$M_{xxzx} = \mathbb{M}_{4c}^{(1)}$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xxyz} & M_{xxyz} & 0 & 0 & 0 & M_{xxzx} \\ -M_{xxzx} & M_{xxzx} & 0 & 0 & 0 & -M_{xxyz} \\ 0 & 0 & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$M_{xxyz} = 2\mathbb{T}_{3c}^{(1)}$
$M_{xxzx} = -2\mathbb{T}_{3s}^{(1)}$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xxyz} & M_{xxzx} & 0 \\ -M_{xxyz} & -M_{xxzx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxzx} & -M_{xxyz} & 0 \end{bmatrix}$$

$M_{xxyz} = \mathbb{T}_{3c}^{(2)}$
$M_{xxzx} = \mathbb{T}_{3s}^{(2)}$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$M_{yzxx} = \mathbb{T}_{3c}^{(3)}$
$M_{yzxy} = \mathbb{T}_{3s}^{(3)}$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xxyz} & M_{xxzx} & 0 & M_{xxzy} & M_{xxxz} & 0 \\ 0 & 0 & 0 & -M_{xxyz} & -M_{xxzx} & 0 & -M_{xxzy} & -M_{xxxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ -M_{yzxy} & M_{yzxy} & 0 & 0 & 0 & M_{yzxx} & 0 & 0 & M_{yzxx} \\ 0 & 0 & 0 & -M_{xxxz} & M_{xxzy} & 0 & -M_{xxzx} & M_{xxyz} & 0 \end{bmatrix}$$

$$\begin{aligned} M_{xxyz} &= \mathbb{M}_{4s}^{(1)} + 2\mathbb{T}_{3c}^{(1)} + \mathbb{T}_{3c}^{(2)} \\ M_{xxzx} &= \mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} + \mathbb{T}_{3s}^{(2)} \\ M_{xxzy} &= \mathbb{M}_{4s}^{(1)} + 2\mathbb{T}_{3c}^{(1)} - \mathbb{T}_{3c}^{(2)} \\ M_{xxxz} &= \mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} - \mathbb{T}_{3s}^{(2)} \\ M_{yzxx} &= \mathbb{M}_{4s}^{(1)} - 2\mathbb{T}_{3c}^{(1)} \\ M_{yzxy} &= -\mathbb{M}_{4c}^{(1)} - 2\mathbb{T}_{3s}^{(1)} \end{aligned}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\begin{aligned} M_{yzxx} &= \mathbb{T}_{3c}^{(3)} \\ M_{yzxy} &= \mathbb{T}_{3s}^{(3)} \end{aligned}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxxz} \\ 0 & 0 & M_{yyyz} \\ 0 & 0 & 0 \\ -M_{xxxz} & M_{yyyz} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M_{yyyz} \\ 0 & 0 & 0 \\ M_{xxxz} & -M_{yyyz} & 0 \\ 0 & 0 & -M_{xxxz} \\ -M_{yyyz} & -M_{xxxz} & 0 \end{bmatrix}$$

$$\begin{aligned} M_{xxxz} &= \mathbb{M}_{4c}^{(1)} \\ M_{yyyz} &= -\mathbb{M}_{4s}^{(1)} \end{aligned}$$